



Problem Challenge 1

We'll cover the following



- Tree Diameter (medium)
- Try it yourself

Tree Diameter (medium)

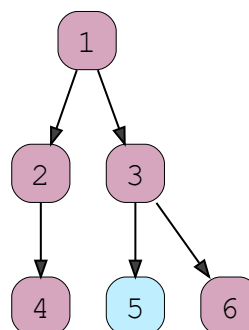
Given a binary tree, find the length of its diameter. The diameter of a tree is the number of nodes on the **longest path between any two leaf nodes**. The diameter of a tree may or may not pass through the root.

Note: You can always assume that there are at least two leaf nodes in the given tree.

Example 1:

Output: 5

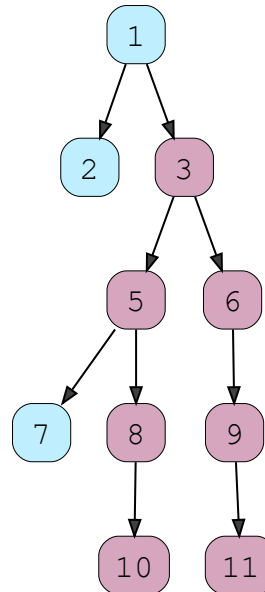
Explanation: The diameter of the tree is: [4, 2, 1, 3, 6]



Example 2:

Output: 7

Explanation: The diameter of the tree is: [10, 8, 5, 3, 6, 9, 11]

**Try it yourself**

Try solving this question here:



Java



Python3



JS



C++

```

1  class TreeNode {
2      int val;
3      TreeNode left;
4      TreeNode right;
5
6      TreeNode(int x) {
7          val = x;
8      }
9  };
10
11 class TreeDiameter {
12
13     public static int findDiameter(TreeNode root) {
14         // TODO: Write your code here
15         return -1;
16     }
17
18     public static void main(String[] args) {
19         TreeNode root = new TreeNode(1);
20         root.left = new TreeNode(2);
21         root.right = new TreeNode(3);
22         root.left.left = new TreeNode(4);
23         root.right.left = new TreeNode(5);
24         root.right.right = new TreeNode(6);
25         System.out.println("Tree Diameter: " + TreeDiameter.findDiameter(root));
26         root.left.left = null;
27         root.right.left.left = new TreeNode(7);
28         root.right.left.right = new TreeNode(8);
29         root.right.right.left = new TreeNode(9);
30         root.right.left.right.left = new TreeNode(10);
31         root.right.right.left.left = new TreeNode(11);
32         System.out.println("Tree Diameter: " + TreeDiameter.findDiameter(root));
  
```

